F7AB - Actuarial and Financial Mathematics

Tutorial 8 – Fixed Interest Bonds, Tax and Index-Linked Securities

Question 1 A government has just issued a fixed interest security (or bond) of nominal amount £100 pays a coupon of 6% p.a. at the end of each of the next ten years and is redeemed at par at the end of the ten years.

1. Determine the price of the security which gives a redemption yield of 6% p.a. effective to a purchaser who pays no tax. Explain the answer.

2. Calculate the price of the security that gives a redemption yield to the same purchaser of 8% p.a. effective.

Question 2 Suppose that in the above example the coupon is £6 p.a. payable half-yearly.

1. Calculate the price of the security that gives a yield to the same purchaser of: (a) 6% p.a. effective, and (b) 8% p.a. effective.

2. Explain why your result in (a) is not £100.

Question 3 A fixed interest security of nominal amount £100 000 will be repaid at 110% after 15 years. The security bears a coupon of 9% p.a. payable annually in arrears. On the issue date, the security is purchased for £80 000 by an investor who is liable to income tax at 40% and to capital gains tax at 30%.

1. Calculate the investor's net (meaning "after tax") effective redemption yield p.a. for the transaction.

2. The investor wishes to secure an after-tax redemption yield of 9% p.a. Determine how much the investor should pay for the security.

Question 4 Consider a 20-year bond, bearing a half-yearly coupon of 8% per annum and redeemable at 120%. It is exactly six months until the next coupon. Calculate the price of the bond to give a redemption yield of 9% per annum effective after tax to an investor who pays:

1. no taxes.

2. income tax at 20% but no capital gains tax (CGT).

3. income tax at 20% and CGT at 30%.

Question 5 On 1 January 2003, a corporation issues a number of bonds, each of nominal amount £100, to be redeemed at 98% on 1 January 2028. The coupon is 7% per annum, payable half-yearly on each 1 January and 1 July.

1. The bond is priced at issue to give a redemption yield of 8.16% per annum to a non-tax-paying investor. Calculate the issue price per £100 nominal.

2. Anna, who pays income tax at 25% and CGT at 40%, bought some of the bonds on the issue date. On 1 July 2013, just after the coupon then due was paid, she sold the bonds to Bob, who pays both income tax and CGT at 40%. Bob paid a price that gave him a redemption yield after taxes of 6% p.a. effective. Calculate the price that Bob paid per £100 nominal.

3. Calculate the yield realised by Anna on her completed transaction.

Question 6 Investors often allow for risk by requiring a higher yield to lend money to a borrower who is thought to be at higher risk of not repaying some or all of the interest and capital. Explain the consequences for companies or governments who are trying to raise money in the bond markets.

Question 7 An investor buys a 20-year bond, to be redeemed at 110% and bearing semi-annual coupons at rate 6% per annum, to obtain a redemption yield of 5% p.a. effective. The investor pays income tax at 25% but no CGT.

1. Calculate the price paid by the investor.

2. Just after the 20th coupon payment, the investor's rate of income tax drops to 20%. If she holds the bond to redemption, calculate her realised yield on the whole transaction.

Question 8 A fixed interest stock with nominal value of £100 was issued on 1 January 1995. The coupon rate was 4% p.a. payable half-yearly in arrears for the first 10 years and then 3% p.a. payable half-yearly in arrears for the next 10 years. Redemption was at par after 20 years. On 1 July 1999, just after the July dividend was paid, the stock was quoted at a price which would have given a tax-free investor a redemption yield of 6% p.a. The stock was bought on that date by an investor who was subject to income tax at 40% and to capital gains tax at 20%.

1. The investor held the bond until the redemption date. Calculate his after tax redemption yield.

2. On 1 August 2000 the stock was priced to give an after tax redemption yield of 5% p.a. effective to the taxed investor described above. Calculate the price of the stock on 1 August 2000. (Note: be careful with the dates.)

Question 9 State the characteristics of index-linked government bonds.

Question 10 On 9 October 2007 an investor, who is not liable to tax, had the choice of purchasing either:

(A) $7\frac{1}{2}\%$ Treasury Stock at price £107 per £100 nominal repayable at par.

(B) 2% Index-Linked Treasury Stock 2016 at a price of £207 per £100 nominal.

The RPI base figure for indexing is 69.5 and the next coupon is to be paid on 9 April 2008. The time lag for inflation indexing is 8 months.

Both stocks pay interest half-yearly in arrears and are redeemed on 9 October 2016. The latest published value of the RPI is 207.3, which is the RPI for August 2007.

Assuming that today is 9 October 2007 and the RPI will grow continuously at a rate of 2.5% per annum from its latest published value,

1. Verify that the real yield from stock (A) as at 9 October 2007 is 4% per annum effective.

2. By considering the series of future receipts from stock (B), derive a formula for the present real value of £100 nominal of that stock.

3. Calculate the real yield as at 9 October 2007 from stock (B) by interpolating between two suitable interest rates which have a difference of at most 0.5%.

Question 11 On 15 March 2006 the government of a country issued an index-linked bond of term 6 years. Coupons are payable half-yearly in arrears, and the annual nominal coupon rate is 3%.

Interest and capital payments are indexed by reference to the value of an inflation index with a time lag of 8 months.

A tax-exempt investor purchased the stock at £111 per £100 nominal on 16 September 2009, just after the coupon payment had been made.

You are given the following values of the inflation index:

Date	Jul 2005	Mar 2006	Jul 2009	Sep 2009
Inflation Index (Date)	110.5	112.1	126.7	127.4

1. Calculate the amount of the coupon payment per £100 nominal stock on 15 March 2010.

2. Calculate the effective real annual yield to the investor on 16 September 2009. You should assume that the inflation index will increase at a constant rate of 4% per annum effective from its value in September 2009.

3. Without doing any further calculations, explain how the effective real annual yield would alter, if at all, if the inflation index for July 2005 had been more than 110.5 and all the other index values were unchanged.

Answers: Q1.1. £100; Q2.1. £86.57984; Q2.1(a). £100.65278; Q2.1(b). £87.37017; Q3.1. 0.0772; Q3.2. £70.47126; Q4.1. £96.04715; Q4.2. £81.12001; Q4.3. £78.92111; Q5.1. £88.97748; Q5.2. £79.42754; Q5.3. 5.12% p.a.; Q7.1. £98.23044; Q7.2. 5.001% p.a.; Q8.1. 3.999% p.a.; Q8.2. £67.17545; Q10.3. 6.481%; Q11.1. £1.71991; Q11.2. 3.52% p.a.

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Solutions to Tutorial 8 – Fixed Interest Bonds, Tax and Index-Linked Securities

Solution 1

1. The price is £100. This is because it satisfies the Principle of Equivalence:

$$100 = 6\, a_{\overline{10}|}^{0.06} + 100 \times 1.06^{-10}.$$

2. If the desired yield is 8% p.a. effective, we must solve:

$$P = 6\, a_{\overline{10}|}^{0.08} + 100 \times 1.08^{-10} \Rightarrow P = £86.57984.$$

Why would we not round the answer to, e.g. 2 d.p.s? Prices of bonds are usually expressed as the price of buying £100 nominal of the bond. However, investors would buy many more pounds nominal of the bond than £100. Usually, investors buy nominal amounts in excess of £10 000 nominal.

Solution 2

1. Working in time units of half-years, we solve

$$P = \frac{6}{2}\, a_{\overline{20}|}^{0.029563} + 100 \times 1.029563^{-20} \quad \Rightarrow \quad P = £100.65278.$$

This is greater than £100 (as in Q.1) because part of the annual interest payment of £6 p.a. is received slightly earlier, so the bond is worth more.

2. Working in time units of half-years, we solve

$$P = 3\, a_{\overline{20}|}^{0.039230} + 100 \times 1.039230^{-20} \quad \Rightarrow \quad P = £87.37017.$$

Solution 3

1. Capital Gains Tax (CGT) is payable, so the equation of value (per £100 nominal) is:

$$\begin{aligned} f(i) &:= -80\,(1+i)^{15} + 0.6 \times 9 \times s_{\overline{15}|}^{i} + 110 - 0.3(110-80) = 0 \\ &\Rightarrow -80\,(1+i)^{15} + 5.4\, s_{\overline{15}|}^{i} + 101 = 0. \end{aligned}$$

We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error:

$$\begin{aligned} f(0.07) &= +15.974 \\ f(0.08) &= -6.152 \\ \Rightarrow i^\star &\approx 0.07 + 0.01\, \frac{15.974}{15.974 + 6.152} = 0.0772. \end{aligned}$$

2. Since $0.09 > 0.0772$ we know the price paid will be less than £80 000 so CGT will certainly be payable. Therefore by the Principle of Equivalence:

$$\begin{aligned} P &= 5.4 a_{\overline{15}|}^{0.09} + (110 - 0.3(110 - P))\, 1.09^{-15} \\ \Rightarrow P(1 - 0.3 \times 1.09^{-15}) &= 5.4 a_{\overline{15}|}^{0.09} + 77 \times 1.09^{-15} \\ \Rightarrow P &= £70.47126. \end{aligned}$$

Solution 4 In each case we switch to a time unit of a half-year, and a redemption yield of 0.044031 per half-year effective. Per £100 nominal:

1. $P = 4\, a_{\overline{40}|}^{0.044031} + 120 \times 1.044031^{-40} \Rightarrow P = £96.04715.$
2. $P = 3.2\, a_{\overline{40}|}^{0.044031} + 120 \times 1.044031^{-40} \Rightarrow P = £81.12001.$
3. $(1 - t_1)\, d/R = 0.02667 < 0.044031$ so CGT is payable.

$$\begin{aligned} P &= 3.2\, a_{\overline{40}|}^{0.044031} + (120 - 0.3(120 - P)) \times 1.044031^{-40} \\ \Rightarrow P(1 - 0.3 \times 1.044031^{-40}) &= 3.2\, a_{\overline{40}|}^{0.044031} + 84 \times 1.044031^{-40} \\ \Rightarrow P &= £78.92111. \end{aligned}$$

Solution 5

1. The term is 25 years. We use a time unit of half a year, and a yield of 0.04 per half-year effective.
$$P = 3.5\, a_{\overline{50}|}^{0.04} + 98 \times 1.04^{-50} \Rightarrow P = £88.97748.$$
2. On 1 July 2013 the bond has 14.5 years (29 half-years) to run. Bob's effective yield is 0.029563 per half-year. From Bob's tax position $(1 - t_1)\, d/R = 0.02143 < 0.029563$ so Bob has to pay CGT.
$$\begin{aligned} P &= 2.1\, a_{\overline{29}|}^{0.029563} + (98 - 0.4(98 - P)) \times 1.029563^{-29} \\ \Rightarrow P(1 - 0.4 \times 1.029563^{-29}) &= 2.1\, a_{\overline{29}|}^{0.029563} + 58.8 \times 1.029563^{-29} \\ \Rightarrow P &= £79.42754. \end{aligned}$$
Anna's financial and tax position are irrelevant to Bob's calculation.
3. Anna sold the bonds for less than the price paid for them so does not pay any CGT. She had held the bonds for 10.5 years. The equation of value, based on Anna's tax position and a time unit of half a year, is
$$f(i) := -88.97748 \times (1 + i)^{21} + 0.75 \times 3.5 \times s_{\overline{21}|}^{i} + 79.42754 = 0.$$
However, to make it easier to plug the equation of value into our calculator, we round the buying and selling prices to 2 d.p.s and set
$$f(i) := -88.98 \times (1 + i)^{21} + 0.75 \times 3.5 \times s_{\overline{21}|}^{i} + 79.43 = 0.$$
We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error (we know that Anna's yield must be less than about $0.75 \times 3.5/88.98 \approx 3\%$ per half year):
$$\begin{aligned} f(0.02) &= +12.247 \\ f(0.03) &= -10.823 \\ \Rightarrow i^\star &\approx 0.02 + 0.01\, \frac{12.247}{12.247 + 10.823} = 0.0253. \end{aligned}$$
Hence Anna's effective annual yield is approximately 5.12%.

Solution 6 Yields are inversely related to price, so for a fixed set of future cashflows, the higher the yield, the lower the price today of the future cashflows. By demanding a higher yield from borrowers thought to be at greater risk of defaulting, lenders make it more expensive for high-risk borrowers to raise money (a company or government who issues a bond is borrowing money). From the lender's point of view they require a higher reward for taking on the risk of the issuing company defaulting on the interest and capital repayments, an example of the relationship between risk and reward that underlies most of financial mathematics.

Solution 7

1. Using a time unit of half a year and an effective yield of 0.024695 per half year:

$$P = 2.25\, a_{\overline{40}|}^{0.024695} + 110 \times 1.024695^{-40} \Rightarrow P = £98.23044.$$

2. The equation of value can be written as (where, as before, we round the price P to 2 d.p.s for ease of calculation)

$$f(i) := -98.23\,(1+i)^{40} + 2.25\, s_{\overline{20}|}^{i} \times (1+i)^{20} + 2.4 s_{\overline{20}|}^{i} + 110 = 0.$$

As the coupon payments are relatively small and the change in the tax rate is not too large, we do not expect a large change in the yield. We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error, in time units of half a year:

$$\begin{aligned} f(0.02) &= -32.653 \\ f(0.03) &= +36.747 \\ \Rightarrow i^\star &\approx 0.02 + 0.01\,\frac{32.653}{32.653 + 36.747} = 0.0247. \end{aligned}$$

Hence the annual yield is approximately 5.001% effective. (Note: this answer would be acceptably accurate in an exam, however you should spot that it is unlikely that a change amounting to 5% of half the coupons paid should have such a small effect on the yield. Interpolation between $i = 0.025$ and $i = 0.026$ gives a more accurate yield of about 5.10% per annum effective.)

Solution 8

1. On 1 July 1999 the 9th coupon has just been paid, and the outstanding term is 15.5 years (31 half years). Using a time unit of half a year, the price on that date was:

$$P = 1.5\, a_{\overline{31}|}^{0.029563} + 0.5\, a_{\overline{11}|}^{0.029563} + 100 \times 1.029563^{-31} \Rightarrow P = £75.3413 \qquad (75.34 \text{ rounded}).$$

On redemption, CGT was payable. The CGT is $20\% \times (100 - P) = £4.93174$. After payment of the CGT from the redemption proceeds, the investor is left with £95.06826.

The equation of value, in half-year time units, can be written as:

$$f(i) = -75.34\,(1+i)^{31} + 1.2\, s_{\overline{31}|}^{i} - 0.3\, s_{\overline{20}|}^{i} + 95.07 = 0.$$

We know this investor must get a lower yield than the yield that the tax-free investor obtained. We use linear interpolation to find the solution $i^\star$ to the equation of value. By trial and error, in time units of half a year:

$$\begin{aligned} f(0.01) &= +29.261 \\ f(0.02) &= -0.561 \\ \Rightarrow i^\star &\approx 0.01 + 0.01\,\frac{29.261}{29.261 + 0.561} = 0.0198. \end{aligned}$$

Hence the annual yield is approximately 3.999% effective.

2. On 1 August 2000, the outstanding term is 14 years and 5 months, or 28.833 half-years, and 29 coupons remain to be paid. The test for CGT to be payable does not work exactly since the coupon is not level, but using the higher level of coupon payment, $(1-t_1)\,d/R = 0.012 < 0.024695$ so we may be confident that CGT is payable. Then the price is the price just after payment of the coupon paid on 1 July 2000, accumulated for 1 month, which can be written:

$$\begin{aligned} P &= 1.024695^{1/6}(0.9\,a^{i}_{\overline{29}|} + 0.3\,a^{i}_{\overline{9}|} + (100 - 0.2(100-P))\,(1.024695)^{-29}) \\ &\Rightarrow \frac{P(1-0.2\times(1.024695)^{-29})}{1.024695^{1/6}} = 0.9\,a^{i}_{\overline{29}|} + 0.3\,a^{i}_{\overline{9}|} + 80(1.024695)^{-29} \\ &\Rightarrow P = £67.17545. \end{aligned}$$

Solution 9

1. Payments guaranteed by government.
2. Low default risk.
3. Can be various different indexation provisions but, in general, protection is given against a fall in the purchasing power of money.
4. Fairly liquid (i.e. large issue size and ability to deal in large quantities) compared with corporate issues, but not compared with conventional issues.
5. Normally coupon and capital payments both indexed to increases in a given price index with a lag.
6. Low volatility of return and low expected real return.
7. More or less guaranteed real return if held to maturity (can vary due to indexation lag).
8. Money return is not guaranteed.

Solution 10

1. As the inflation index is assumed to grow at a constant rate of 2.5% p.a. we can use the fact that the real yield i' and the money yield i are related by

$$1 + i' = \frac{1+i}{1.025}.$$

to do the verification. If the real yield i'_A for stock A is 4% p.a. then the money yield of A must be $i_A = (1+i'_A)(1.025) - 1 = 0.066$ p.a. As

$$107 - 7.5a^{(2)}_{\overline{9}|} - 100v^9 = -0.22814 \quad \text{at } i_A = 0.066,$$

this means that the money yield is 6.6% p.a, which verifies that the real yield is 4% p.a.

2. As the RPI at the base date is 69.5, we estimate the money payments per £100 nominal from the index-linked stock as follows.

Time t	0.5	1	1.5	$\cdots$	9
Date at time t	9 Apr 08	9 Oct 08	9 Apr 09	$\cdots$	9 Oct 16
Lagged RPI date	Aug 07	Feb 08	Aug 08	$\cdots$	Feb 16
RPI(Lagged date)	207.3	$207.3\times1.025^{\frac{1}{2}}$	207.3×1.025^{1}	$\cdots$	$207.3\times1.025^{8\frac{1}{2}}$
Money payment	$\frac{207.3}{69.5}$	$\frac{207.3\times1.025^{\frac{1}{2}}}{69.5}$	$\frac{207.3\times1.025^{1}}{69.5}$	$\cdots$	$101\times\frac{207.3\times1.025^{8\frac{1}{2}}}{69.5}$

The real values calculated at 9 October 2007 can be found as follows. As the projected RPI at October 2007 is $207.3 \times 1.025^{\frac{2}{12}}$,

Time t	0.5	1	1.5	$\cdots$
Money payment	$\frac{207.3}{69.5}$	$\frac{207.3 \times 1.025^{\frac{1}{2}}}{69.5}$	$\frac{207.3 \times 1.025^{1}}{69.5}$	$\cdots$
Payment date	9 Apr 08	9 Oct 08	9 Apr 09	$\cdots$
RPI(Payment date)	$207.3 \times 1.025^{\frac{8}{12}}$	$207.3 \times 1.025^{1\frac{2}{12}}$	$207.3 \times 1.025^{1\frac{8}{12}}$	$\cdots$
Real payment	$\frac{207.3}{69.5} \times \frac{207.3 \times 1.025^{\frac{2}{12}}}{207.3 \times 1.025^{\frac{8}{12}}}$	$\frac{207.3 \times 1.025^{\frac{1}{2}}}{69.5} \times \frac{207.3 \times 1.025^{\frac{2}{12}}}{207.3 \times 1.025^{1\frac{2}{12}}}$	$\frac{207.3 \times 1.025^{1}}{69.5} \times \frac{207.3 \times 1.025^{\frac{2}{12}}}{207.3 \times 1.025^{1\frac{8}{12}}}$	$\cdots$
at Oct 2007	$= \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$	$= \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$	$= \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$	$\cdots$

with the total real value of the redemption payment and last coupon payment being $101 \times \frac{207.3}{69.5} \times \frac{1}{1.025^{\frac{1}{2}}}$.

Hence if i'_B is the real rate of return, the formula for the present real value of £100 nominal of stock (B) is:

$$\frac{207.3}{69.5(1.025)^{1/2}} \left(2a_{\overline{9}|}^{(2)i'_B} + 100(1+i'_B)^{-9}\right). \qquad (1)$$

3. Having found the expression (1), the real yield is most easily found by solving the equation of value formulated at $t = 0$, that is, on 9 October 2007. We express it as follows:

$$207 = \frac{207.3}{69.5(1.025)^{1/2}} \left(2a_{\overline{9}|}^{(2)i'_B} + 100(1+i'_B)^{-9}\right).$$

By trial and error, we find the solution lies between 0.06 and 0.065. Then linear interpolation gives $i'_B \approx 0.06481$.

Solution 11

1. As the base date for indexing is July 2005, the coupon will be:

$$\frac{0.03 \times 100}{2} \times \frac{\text{II(Jul 2009)}}{\text{II(Jul 2005)}} = 1.5 \times \frac{126.7}{110.5} = £1.71991 \text{ per £100 nominal.}$$

2. As the RPI at the base date is 110.5, we estimate the money payments per £100 nominal from the index-linked stock as follows.

Time t	0.5	1	1.5	2	2.5
Date at time t	Mar 10	Sep 10	Mar 11	Sep 11	Mar 12
Lagged RPI date	Jul 09	Jan 10	Jul 10	Jan 11	Jul 11
RPI(Lagged date)	126.7	$127.4 \times 1.04^{\frac{4}{12}}$	$127.4 \times 1.04^{\frac{10}{12}}$	$127.4 \times 1.04^{1\frac{4}{12}}$	$127.4 \times 1.04^{1\frac{10}{12}}$
Money payment	$1.5\frac{126.7}{110.5}$	$1.5\frac{127.4 \times 1.04^{\frac{4}{12}}}{110.5}$	$1.5\frac{127.4 \times 1.04^{\frac{10}{12}}}{110.5}$	$1.5\frac{127.4 \times 1.04^{1\frac{4}{12}}}{110.5}$	$101.5\frac{127.4 \times 1.04^{1\frac{10}{12}}}{110.5}$

The real values are calculated at September 2009, at which date the RPI is 127.4.

Time t	0.5	1	1.5	$\cdots$
Money payment	$1.5\frac{126.7}{110.5}$	$1.5\frac{127.4\times1.04^{\frac{4}{12}}}{110.5}$	$1.5\frac{127.4\times1.04^{\frac{10}{12}}}{110.5}$	$\cdots$
Payment date	Mar 10	Sep 10	Mar 11	$\cdots$
RPI(Payment date)	$127.4(1.04)^{0.5}$	$127.4(1.04)^{1}$	$127.4(1.04)^{1.5}$	$\cdots$
Real payment	$1.5\frac{126.7}{110.5}\frac{127.4}{127.4(1.04)^{0.5}}$	$1.5\frac{127.4\times1.04^{\frac{4}{12}}}{110.5}\frac{127.4}{127.4(1.04)^{1}}$	$1.5\frac{127.4\times1.04^{\frac{10}{12}}}{110.5}\frac{127.4}{127.4(1.04)^{1.5}}$	$\cdots$
at Sep 2009	$=1.5\frac{126.7}{110.5}\frac{1}{(1.04)^{0.5}}$	$=1.5\frac{127.4}{110.5}\frac{1}{(1.04)^{\frac{8}{12}}}$	$=1.5\frac{127.4}{110.5}\frac{1}{(1.04)^{\frac{8}{12}}}$	$\cdots$
	$=1.68651$	$=1.68478$	$=1.68478$	$\cdots$

with the total real value of the redemption payment and last coupon payment being

$$101.5\frac{127.4}{110.5}\frac{1}{(1.04)^{\frac{8}{12}}}=114.00336.$$

Using time units equal to 6 months, the equation of real value is

$$f(j):=111(1+j)^5-1.68651(1+j)^4-1.68478(1+j)^3-1.68478(1+j)^2-1.68478(1+j)^1-114.00336.$$

As a first estimate of the solution to $f(j^\star)=0$, assume that the all the income is received at the end of the investment. Then $111(1+j)^5\approx120.74421$, resulting in a first estimate of $j^\star\approx1.7\%$.

As $f(0.015)=-1.422$ and $f(0.02)=1.465$, linear interpolation gives $j^\star\approx1.746\%$. Hence the annual real yield is 3.52%.

3. The real yield decreases. If the inflation index at July 2005 had been greater than 110.5, then the money payments would decrease. Consequently, so would the real payments. Thus the investor would be paying the same amount £111 for less and hence the investor's yield would be less than 3.52% p.a.